FILED
05/29/26
04:59 PM
R2508004

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**

Order Instituting Rulemaking to Update Distribution Level Interconnection Rules and Regulations.	Rulemaking 25-08-004 (Filed August 14, 2025)

REPLY COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE ASSIGNED COMMISSIONER'S SCOPING MEMO AND RULING

Zach Woogen
Executive Director
Vehicle-Grid Integration Council
1401 21st St., Suite 5409
Sacramento, California 95811
Telephone: (279) 895-7855
Email: vgicregulatory@vgicouncil.org

Grace Pratt
Caliber Strategies
801 K Street, Suite 2800
Sacramento, CA 95814
Telephone: (818) 264-6437
Email: grace@caliberstrat.com

Date: May 29, 2026

BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA

Order Instituting Rulemaking to Update Distribution Level Interconnection Rules and Regulations.	Rulemaking 25-08-004 (Filed August 14, 2025)

REPLY COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE ASSIGNED COMMISSIONER'S SCOPING MEMO AND RULING

In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission ("Commission"), the Vehicle-Grid Integration Council ("VGIC") hereby submits these reply comments on the *Assigned Commissioner's Scoping Memo and Ruling* ("Scoping Memo"), issued by President Alice Reynolds on March 3, 2026, as modified by the *Email Ruling Granting Extension for Comments, Motion to Late File NOI, and Motions for Party Status* issued by Administrative Law Judge Andrew Dugowson.

I. INTRODUCTION.

VGIC appreciates opening comments from parties on improvements to Rule 21 and Phase 1 issues. As California continues to advance transportation electrification, updates to Rule 21 will play an increasingly important role in enabling bidirectional electric vehicle ("EV") charging systems to meet grid needs.

In this proceeding, VGIC looks forward to further improvements to Rule 21 that will better enable the interconnection of distributed energy resources ("DERs") and bidirectional charging systems. In Opening Comments on the Scoping Memo, parties broadly discussed opportunities to streamline interconnection processes for simpler projects, particularly non-export systems.[1] VGIC

[1] Advanced Energy United Opening Comments at 2; Electrify America Opening Comments at 4-6; PG&E Opening Comments at 17; Tesla Opening Comments at 5-12.

supported these efforts in comments on the Order Instituting Rulemaking and encourages the Commission to continue prioritizing streamlined review pathways and updates to technical requirements that can improve interconnection efficiency while maintaining safety and reliability standards. This effort may also include further clarification and refinement of bidirectional charging system configurations and operational definitions. To the extent these issues cannot be fully addressed during this phase of the proceeding, the Commission should prioritize more detailed discussion of interconnection streamlining and technical requirement updates in Phase 2.

In Phase 1 of this proceeding, the Commission should ensure that interconnection processes and fees appropriately reflect the operational characteristics and grid impacts of these technologies and avoid creating unnecessary barriers to deployment. In these reply comments, VGIC provides the following comments:

- Small non-NEM bidirectional charging systems should use the existing NEM / NBT interconnection fee.
- The Commission should clarify that isolated, backup-only bidirectional systems and load-only EV systems with disabled grid-parallel features[2] are not subject to Rule 21 interconnection requirements.

II. SMALL NON-NEM BIDIRECTIONAL CHARGING SYSTEMS SHOULD USE THE EXISTING NEM / NBT INTERCONNECTION FEE.

In Opening Comments, VGIC recommended that the Commission set the interconnection application fee for non-NEM bidirectional charging systems equal to the interconnection fee

[2] Load-only EV systems with disabled grid-parallel features are defined as systems that have the hardware necessary to operate in parallel with the grid, but they cannot and do not operate in parallel with the grid because they are prevented from doing so due to software controls and/or software locks. These systems operate solely as load-only, EV charging resources.

currently in place for small NEM systems. Other parties made similar recommendations, with Tesla and CALSSA explicitly recommending that the non-NEM interconnection application fee align with the current NEM amounts.

These recommendations were made largely because of similarities between small NEM and non-NEM systems. Tesla notes that "a residential non-exporting solar-plus-storage addition, a residential standalone storage system or a V2G interconnection application has more in common and probably should be directed to the same group and leverage the same review processes as NEM/NBT projects."[3] CALSSA similarly notes that, "Such projects are not technically more complex and do not create any novel risks to the grid. Many of the same processes used to evaluate residential-scale NEM/NBT systems can be applied to evaluate smaller-scale V2G and exporting standalone storage systems."[4] SEIA also proposes to eliminate the differentiation between NEM and non-NEM projects and agrees that "Given the comparability of those projects both in terms of scale/complexity and potential grid impacts disparate fees are not warranted."[5]

Currently, grid-parallel, bidirectional charging systems are treated under Rule 21 as a form of energy storage because they use the same battery technology as stationary energy storage. Namely, bidirectional charging systems use UL 1741 SB-certified systems: 1) The Ford Charge Station Pro + Home Integration System, 2) GM Energy PowerShift Charge + V2H Enablement Kit, 3) dcbel Ara, 4) Wallbox Quasar 2, 5) Nuvve TP-60DC-V2G-H1-C, Nuvve TP-40DCV2G-H1-C, and Nuvve TP-20DC-V2G-H1-C, 6) InCharge Energy ICE-22-V2X, InCharge Energy ICE44-V2X, InCharge Energy ICE-66 V2X, and InCharge Energy ICE-132 V2X, 7) Tellus Power TP-V2G-60-480, Tellus Power TP-V2G-40-480, Tellus Power TP-V2G-30-480, and Tellus Power

[3] Tesla Opening Comments at 15.
[4] CALSSA Opening Comments at 19.
[5] SEIA Opening Comments at 65.

TP-V2G-20-480 bidirectional charging systems are all certified to UL 1741 SB.[6] In addition, the Tesla Powershare Grid Support system, leveraging the Tesla Universal Wall Connector, now meets IEEE 1547-2018 via a new UL 1741 SB Certification Requirements Decision ("CRD") in April 2026.[7] These systems are not only certified to the same standards as stationary storage devices, but can even be the exact same models, with companies like Enphase using the same inverter models for their residential stationary storage and forthcoming bidirectional charger products.[8] These certifications and inverter similarities ensure that bidirectional charging systems operate with the same safety and reliability features that utilities are already very familiar with from their processing of stationary storage and solar interconnections.

Given the similarities between these products, utilities should require similar levels of review effort and associated costs for interconnection applications for NEM / NBT systems, non-NEM/NBT stationary storage, and grid-parallel bidirectional charging systems. The Commission should therefore adopt a fee structure for small bidirectional charging systems that aligns with existing NEM interconnection application fees and directs utilities to process these projects through comparable residential storage review pathways.

In Opening Comments, VGIC recommended that, if the Commission wishes to set the interconnection fee for bidirectional charging systems based on the costs of installing those systems, it should do so only after reaching a specified threshold. Specifically, VGIC

---

[6] See: California Energy Commission V2G Equipment List https://v2gel.energy.ca.gov/Home/ProcessView. Accessed April 13, 2026; GM Energy Inverter Specifications. https://gmenergy.gm.com/content/dam/gmenergy/na/us/en/index/01-pdfs/gm-energy-inverter-specifications.pdf

[7] UL, Certificate Performance of Grid Interactive Equipment and Systems for Tesla, Inc.: https://energylibrary.tesla.com/docs/Public/More/ProductCertificationInformation/Powershare_UL1741_VoC.pdf

[8] Enphase uses IQ microinverters for their stationary storage and bidirectional products. See the IQ Bidirectional EV Charger specifications: https://enphase.com/download/iq-bidirectional-ev-charger-data-sheet; IQ Battery 5P specifications: https://enphase.com/en-au/download/iq-battery-5p-anz

recommended that this shift should be triggered after surpassing 10,000 total completed small non-NEM bidirectional charging system interconnection applications across all three IOUs.

However, given the similarities between stationary and mobile storage highlighted by parties in Opening Comments, VGIC modifies its recommendation and instead recommends that any reevaluation of interconnection fees occur after utilities collectively process 10,000 completed small non-NEM *energy storage* interconnection applications across all three IOUs (including non-NEM additions to NEM projects). Because bidirectional charging systems are operationally and technologically like stationary storage systems, the utility experience processing small energy storage interconnection applications is a more appropriate benchmark than experience limited solely to bidirectional EV charging systems.

If interconnection fees are set equally for all types of energy storage, it will also be important for the Commission to evaluate differences in review complexity and cost between projects of different sizes, as highlighted by parties in Opening Comments.[9] The Commission should therefore ensure that any future fee structure appropriately reflects the actual review burdens associated with residential-scale projects and does not impose disproportionate costs on smaller customer-sited systems.

III. THE COMMISSION SHOULD CLARIFY THAT BACKUP-ONLY BIDIRECTIONAL SYSTEMS AND LOAD-ONLY EV SYSTEMS WITH DISABLED GRID-PARALLEL FEATURES ARE NOT SUBJECT TO RULE 21 INTERCONNECTION REQUIREMENTS

In Opening Comments, Ford highlighted the current confusion surrounding the applicability of Rule 21 to systems being used solely for charging and/or backup power. As Ford explains, the Commission has previously discussed that systems operating only for charging and/or

[9] PG&E Opening Comments at 20; SEIA Opening Comments at 64-65; Tesla Opening Comments at 12-13.

backup power in a break-before-make configuration are not intended to proceed through the Rule 21 interconnection process.[10] Instead, backup power customers are generally expected to provide notification to the utility that a backup generator is being installed. However, utilities currently route these notifications through their Rule 21 portals in ways that can be difficult for customers and installers to navigate.

Ford highlights that PG&E's portal requires users to click through the page as if they are submitting a Rule 21 application before selecting a "Standby Emergency Generator" option.[11] SCE similarly requires a Rule 21 application process to be initiated before users can select an "Isolated Backup Power" option.[12] Although these processes appear to function as notification pathways rather than actual Rule 21 applications, their placement within Rule 21 interconnection portals creates unnecessary confusion for customers and installers.

Neither PG&E nor SCE appears to be charging the $800 interconnection application fee for these break-before-make backup power generators, and these projects do not proceed through the Rule 21 evaluation and study process. VGIC commends the utilities for applying this appropriate treatment. However, as Ford notes, the current structure "leaves the possibility for customers and installers of backup power systems to inadvertently bin themselves into categories that would improperly apply the fees and additional processes required by interconnection, which is intended for parallel operation use cases."[13]

---

[10] Ford Opening Comments at 5-6.

[11] Ford Opening Comments at 6.

[12] See SCE Form 14-732: https://edisonintl.sharepoint.com/:b:/t/Public/Misc/ER_UWBusgIlPqlrMiOJLXpAB4djEWq4hq_pVJ7hbWEnsFw?e=RuGxTc

[13] Ford Opening Comments at 7.

This confusion could create unnecessary barriers to the adoption of backup power and bidirectional charging technologies by causing customers and installers to incorrectly believe they must complete full Rule 21 interconnection processes or incur substantial fees for systems intended solely for backup operation during outages. **Clearer guidance and processes would significantly reduce customer uncertainty, minimize administrative burdens, and support broader deployment of electric vehicles and resiliency technologies.**

Although current rules already indicate that load-only EV systems with disabled grid-parallel features and break-before-make, backup-only systems do not need to proceed through Rule 21 interconnection review, VGIC agrees with Ford that **the Commission should explicitly clarify in this proceeding that the following systems are not required to submit Rule 21 interconnection applications or pay Rule 21 interconnection application fees: 1) load-only EV systems that operate as load but have disabled grid-parallel features; and 2) backup generators, including bidirectional charging systems, enabled in bidirectional mode only when isolated from the grid.**

In addition, utilities should revise and clarify their customer-facing forms and processes to ensure that load-only EV systems with disabled grid-parallel features and backup-only systems are <u>clearly</u> distinguished from systems intended to operate in parallel with the grid. Furthermore, VGIC recommends that utilities more explicitly outline backup power notification processes on their websites and either establish separate backup power notification portals or provide a clear "backup power" selection option on the <u>first page</u> and/or <u>initial guidance</u> of the Rule 21 portal. Standardized and transparent processes and guidance across all IOUs would reduce customer confusion and help ensure that backup-only systems are not inadvertently subjected to inappropriate interconnection requirements.

IV. CONCLUSION.

VGIC appreciates the opportunity to provide these comments on the Scoping Memo. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen
Zach Woogen
Executive Director
VEHICLE-GRID INTEGRATION COUNCIL

Date: May 29, 2026